

TROY, AL, USA

WMO: 722267

Lat: 31.861N

Lon: 86.012W

Elev: 397

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 03878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	27.3	9.7	9.2	33.6	14.4	11.6	37.9	20.7	46.7	18.9	47.7	3.2	330	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	95.2	75.2	93.2	75.4	91.4	75.4	79.1	88.7	78.2	87.8	77.5	86.9	6.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	141.6	82.3	75.7	136.5	81.5	75.0	133.1	80.9	42.7	88.5	41.8	87.9	41.0	87.0	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.5	12.8	17.4	97.6	4.0	3.1	14.6	99.9	12.2	101.7	10.0	103.4	7.1	105.7
			16.2	80.9	4.0	1.3	13.3	81.8	10.9	82.6	8.7	83.3	5.8	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	47.4	51.5	57.8	64.7	72.2	78.7	80.6	80.4	76.3	66.0	55.8	50.2
	DBStd	13.85	10.15	9.12	8.64	6.60	5.44	3.15	2.55	3.07	4.92	7.72	8.41	9.43
	HDD50	450	174	85	31	2	0	0	0	0	0	2	35	121
	HDD65	2133	548	386	252	85	13	0	0	0	3	83	296	467
	CDD50	5998	94	127	271	443	687	861	950	943	788	499	209	126
	CDD65	2207	3	9	27	76	235	411	485	478	341	115	20	7
	CDH74	20375	9	41	234	782	2300	3928	4604	4372	2919	1025	137	24
	CDH80	8719	0	2	26	155	876	1872	2192	2056	1248	277	15	0
(14) Wind	WSAvg	4.8	5.9	6.0	6.2	5.3	4.5	3.9	3.6	3.6	4.1	4.4	4.7	5.5
(15) Precipitation	PrecAvg	55.7	4.9	4.8	6.3	4.4	3.6	5.0	6.0	4.5	3.7	2.9	4.2	4.6
	PrecMax	90.9	10.7	11.9	15.4	9.2	10.4	12.6	15.6	11.3	13.3	12.2	10.6	16.3
	PrecMin	40.6	0.9	0.9	0.4	0.7	0.1	1.0	1.7	0.6	0.6	0.0	1.2	1.2
	PrecStd	8.9	2.3	2.6	3.3	2.4	2.1	2.8	3.1	2.4	2.5	2.7	1.9	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	78.7	82.9	86.3	92.5	97.0	96.9	98.7	95.2	89.0	81.9	77.4
		MCWB	65.0	66.5	65.0	68.8	72.0	74.3	75.8	75.7	74.1	73.1	68.3	66.0
	2%	DB	71.4	74.5	79.8	83.9	89.9	94.3	94.4	94.8	92.3	85.9	78.1	73.1
		MCWB	63.4	63.3	64.7	67.6	71.3	74.3	76.1	75.9	74.1	70.2	66.3	65.5
	5%	DB	68.2	70.7	76.5	81.7	87.8	92.2	92.3	92.4	90.1	83.2	74.6	70.0
		MCWB	62.0	62.1	62.9	66.3	70.8	74.6	76.3	76.5	74.0	68.9	64.4	64.2
	10%	DB	64.3	67.6	73.0	79.1	85.3	89.9	90.3	90.2	87.6	80.5	71.9	66.3
		MCWB	59.1	59.6	61.0	64.7	69.7	74.0	76.1	76.2	73.3	67.5	63.3	61.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	69.0	69.3	73.3	75.7	79.3	80.2	80.5	78.3	76.5	73.1	70.5
		MCDB	71.1	74.6	75.5	81.4	86.0	88.9	89.9	89.4	87.6	84.2	77.6	74.0
	2%	WB	65.8	66.7	67.7	71.2	74.3	78.0	78.9	79.3	77.1	74.5	70.6	67.9
		MCDB	69.8	71.7	74.0	78.7	83.9	88.1	88.9	88.4	86.4	80.7	73.9	70.9
	5%	WB	63.4	64.7	66.1	69.5	73.2	77.0	78.1	78.3	76.2	72.9	67.8	65.3
		MCDB	66.4	68.9	73.1	76.6	82.4	87.0	87.9	87.4	85.1	78.5	72.5	68.2
	10%	WB	60.3	62.0	64.0	67.7	72.1	76.0	77.4	77.5	75.3	70.9	65.2	62.5
		MCDB	64.5	66.2	70.5	74.6	81.0	85.2	86.8	86.5	83.6	76.5	70.1	65.9
(35) Mean Daily Temperature Range	5% DB	MDBR	22.4	22.8	23.7	24.6	23.7	20.7	19.5	19.4	20.8	24.4	25.2	21.8
		MCDBR	23.2	24.5	25.6	26.4	26.1	25.0	22.4	22.7	23.9	25.8	26.1	22.3
	5% WB	MCWBR	15.6	14.9	12.6	11.8	10.3	8.4	7.4	7.4	8.6	11.7	15.0	15.4
		MCWBR	19.1	19.7	20.4	19.9	21.4	20.3	19.4	19.1	19.2	18.4	19.3	18.3
(40) Clear-Sky Solar Irradiance	taub		0.322	0.330	0.355	0.383	0.431	0.473	0.523	0.510	0.428	0.369	0.346	0.326
		taud	2.529	2.509	2.439	2.361	2.263	2.203	2.070	2.103	2.317	2.456	2.468	2.531
	Ebn at Noon		287	295	294	288	273	260	247	249	267	277	275	279
		Edh at Noon	27	30	35	39	43	46	52	50	39	31	29	26
(44) All-Sky Solar Radiation	RadAvg		915	1096	1471	1820	1997	1943	1894	1755	1569	1348	1032	789
	RadStd		59	136	127	143	141	152	122	107	125	161	92	85

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	+295	+180
(47) Regional (45 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	+284	+182

Nomenclature: See separate page